

## Problem A. Recall

Input file:            **standard input**  
Output file:          **standard output**  
Time limit:          2 seconds  
Memory limit:        1024 megabytes

There is an initially empty stack  $S$ . A sequence of operations is then performed on it. There are three types of operations:

- `" + x "` means pushing the element  $x$  onto the stack.
- `" ? x "` means querying whether the element  $x$  exists in the stack. There are two possible results:
  - `" T x "` means that the element  $x$  exists.
  - `" F x "` means that the element  $x$  does not exist.
- `" - "` means popping the top element from the stack (the stack must be nonempty).

Little G recorded all `" + x "` `" T x "` and `" F x "` operations during the entire process, but did not record any `" - "` operations.

Now, Little G wants to insert some `" - "` operations into the recorded operation sequence such that:

- At all times, all elements in the stack are distinct.
- The results of all `" ? x "` queries are consistent with the operation sequence you construct.

It is guaranteed that at least one valid reconstructed sequence exists.

## Input

There are multiple test cases.

The first line of the input contains an integer  $T$  ( $1 \leq T \leq 10^5$ ), representing the number of test cases. Each test case is described as follows:

For each test case:

- The first line contains a positive integer  $n$  ( $1 \leq n \leq 10^6$ ).
- Each of the next  $n$  lines is in the form `" + x "` `" T x "` or `" F x "` representing a recorded operation. It is guaranteed that  $1 \leq x \leq 10^9$ .

It is guaranteed that  $1 \leq \sum n \leq 10^6$ .

## Output

For each test case, output one line containing a string consisting only of the characters `" + "` `" ? "` and `" - "` representing the operation sequence you reconstruct, where:

- If the  $i$ -th character is `" + "` then the  $i$ -th operation in your reconstructed sequence pushes an element onto the stack.
- If the  $i$ -th character is `" ? "` then the  $i$ -th operation in your reconstructed sequence queries whether an element exists in the stack.
- If the  $i$ -th character is `" - "` then the  $i$ -th operation in your reconstructed sequence pops the top element from the stack.

If there are multiple possible original operation sequences, you may reconstruct any one of them.

## Example

| standard input | standard output  |
|----------------|------------------|
| 2              | + - + ? - ?      |
| 4              | ++ ? ? ? - - + - |
| + 1            |                  |
| + 1            |                  |
| T 1            |                  |
| F 1            |                  |
| 6              |                  |
| + 1            |                  |
| + 2            |                  |
| T 1            |                  |
| F 3            |                  |
| T 2            |                  |
| + 2            |                  |